# Dr Harin Lee

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## POSITIONS

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|---|---|
| Oct 2025 — | **University of Cambridge**, UK<br>Junior Research Fellow at King's College on the topic of 'Rhythm' |
| Oct 2020 — Sep 2025 | **Max Planck Institute for Human Cognitive and Brain Sciences (MPI CBS)**, Germany<br>PhD in Cognitive Science. Thesis titled 'Global Music Culture' supervised by Dr Marc Schönwiesner & Dr Nori Jacoby — Summa Cum Laude |
| Oct 2020 — Sep 2025 | **International Max Planck Research Schools Fellowship (IMPRS NeuroCom)**, Germany<br>Full-funding by the Max Planck Society on a competitive basis |
| Dec 2020 — Sep 2025 | **Max Planck Institute for Empirical Aesthetics (MPIEA)**, Germany<br>Visiting Researcher at the *Computational Auditory Perception* group led by Dr Nori Jacoby |

## EDUCATION

** Gap between MSc and PhD is due to serving the compulsory Korean military duties for two years*

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|---|---|
| | **Goldsmiths College, University of London**, UK |
| Oct 2017 — Oct 2018 | MSc Music, Mind and Brain — Distinction |
| Oct 2013 — Sep 2017 | BMus Popular Music — First Class |
| | **Richard Huish College**, UK |
| Oct 2011 — Jul 2013 | A-levels: Mathematics, Physics, Music Technology — A* A* A |
| Other | TSE Summer School in Quantitative Social Sciences (2024), Diverse Intelligence Summer Institute Fellow (2022), Neuromatch Academy Summer School (2020) |

## PUBLICATIONS

†Joint First Authorship; *Joint Senior/Last Authorship

Ref. Journal Articles & Archival Conf. Proc

— 2025 —

**Lee, H.**, Jacoby, N., Hennequin, R.*, & Moussallam, M.* (2025). Mechanisms of cultural diversity in urban populations. ***Nature Communications***, 16(1), 5192. https://doi.org/10.1038/s41467-025-60538-2

**Lee, H.**, Çelen, E., Harrison, P. M. C., Anglada-Tort, M., van Rijn, P., Park, M., Schönwiesner, M., & Jacoby, N. (2025). GlobalMood: A cross-cultural benchmark for music emotion recognition. *Proceedings of the International Society for Music Information Retrieval Conference (**ISMIR** 2025)*. https://doi.org/10.48550/arXiv.2505.09539

**Lee, H.**, Van Geert, E., Çelen, E., Marjieh, R., van Rijn, P., Park, M., & Jacoby, N. (2025). Visual and Auditory Aesthetic Preferences Across Cultures. *Proceedings of the Annual Meeting of the Cognitive Science Society (**CogSci**).* https://doi.org/10.48550/arXiv.2502.14439

Çelen, E., van Rijn, P., **Lee, H.***, & Jacoby, N.* (2025). Are Expressions for Music Emotions the Same Across Cultures? *Proceedings of the Annual Meeting of the Cognitive Science Society (**CogSci**).* https://doi.org/10.48550/arXiv.2502.08744

Sguerra, B., Epure, E. V., **Lee, H.**, & Moussallam, M. (2025). Biases in LLM-Generated Musical Taste Profiles for Recommendation. *ACM Recommender Systems (**RecSys**).* https://doi.org/10.1145/3705328.3748030

— 2024 —

Marjieh, R., van Rijn, P., Sucholutsky, I., **Lee, H.**, Griffiths, T. L., & Jacoby, N. (2024). A Rational Analysis of the Speech-to-Song Illusion. *Proceedings of the Annual Meeting of the Cognitive Science Society (**CogSci** 2024).* https://doi.org/10.48550/arXiv.2402.06992

Marjieh, R.[†], Harrison, P. M. C.[†], **Lee, H.**, Deligiannaki, F., Jacoby, N. (2024). Timbral effects on consonance disentangle psychoacoustic mechanisms and suggest perceptual origins for musical scales. ***Nature Communications***, 15(1), 1482. https://doi.org/10.1038/s41467-024-45812-z

— 2023 —

**Lee, H.**, Hennequin, H, & Moussallam, M. (2023). Understanding individual and collective diversity of cultural consumption through large-scale of music listening events. *Computational Humanities Research Conference (**CHR** 2023)*. — Best Paper Award.
https://ceur-ws.org/Vol-3558/paper3539.pdf

Anglada-Tort, M., **Lee, H.**, Krause, A. E., & North, A. C. (2023). Here comes the sun: Music features of popular songs reflect prevailing weather conditions. ***Royal Society Open Science***, 10(5), 221443. https://doi.org/10.1098/rsos.221443

van Rijn, P., Sun, Y., **Lee, H.**, Marjieh, R., Sucholutsky, I., Lanzarini, F., André, E., & Jacoby, N. (2023). Around the world in 60 words: A generative vocabulary test for online research. *Proceedings of the Annual Meeting of the Cognitive Science Society (**CogSci** 2023)*.
https://doi.org/10.1016/j.cub.2023.02.070

Anglada-Tort, M., Harrison, P. M. C., **Lee, H**., & Jacoby, N. (2023). Large-scale iterated singing experiments reveal oral transmission mechanisms underlying music evolution. ***Current Biology***, 0(0). https://doi.org/10.1016/j.cub.2023.02.070

Marjieh, R.[†], van Rijn, P.[†], Sucholutsky, I.[†], Sumers, T., **Lee, H.**, Griffiths, T. L.*, & Jacoby, N.* (2023). Words are all you need? Language as an approximation for human similarity judgments. *International Conference on Learning Representations* (***ICLR** 2023)*.
https://openreview.net/forum?id=O-G91-4cMdv

— 2022 —

van Rijn, P.[†], **Lee, H.**[†], & Jacoby, N. (2022). Bridging the prosody GAP: Genetic Algorithm with People to efficiently sample emotional prosody. *Proceedings of the Annual Meeting of the Cognitive Science Society (**CogSci** 2022)*. https://escholarship.org/uc/item/59q714w6

— 2021 —

**Lee, H.**, Höger, F., Schoenwiesner, M., Park, M., & Jacoby, N. (2021). Cross-cultural mood perception in pop songs and its alignment with mood detection algorithms. *International Society for Music Information Retrieval Conference* (***ISMIR** 2021)*, 366–373. — Finalist for Best Paper Award. https://doi.org/10.5281/zenodo.5625680

— 2020—

**Lee, H**., Launay, J., & Stewart, L. (2020). Signals through music and dance: Perceived social bonds and formidability on collective movement. ***Acta Psychologica***, 208, 103093.
https://doi.org/10.1016/j.actpsy.2020.103093

**Lee, H.**, & Müllensiefen, D. (2020). The Timbre Perception Test (TPT): A new interactive musical assessment tool to measure timbre perception ability. ***Attention, Perception, and Psychophysics***, 82(7), 3658–3675. https://doi.org/10.3758/s13414-020-02058-3

In Review & Preprints

**Lee, H.**[†], Anglada-Tort, M.[†], Sobchuk, O., van Rijn, P., Schönwiesner, M., Tchernichovski, O.*, Park, M.*, & Jacoby, N.* Rapid reshaping of cultural boundaries during war. *In Review.* Preprint
https://doi.org/10.31234/osf.io/7b98u

van Rijn, P., Sun, Y., **Lee, H.**, Marjieh, R., Sucholutsky, I., Lanzarini, F., André, E., Jacoby, N. Evaluating Language Proficiency in 35 languages across 35 countries. *In Review*. Preprint https://doi.org/10.48550/arXiv.2302.01614

Marjieh, R., van Rijn, P., Sucholutsky, I., **Lee, H.**, Jacoby, N., & Griffiths, T. L. (2024). Characterizing the large-scale structure of grounded semantic networks. *Accepted*. Preprint https://doi.org/10.31234/osf.io/phte5

## PROFESSIONAL EXPERIENCES & ACTIVITIES

Research Expedition
Jul 2025

**Field Experiments**, Bolivia
*Conducted experiments across small-scale society Tsimané villages in the Bolivian Amazon* — video documentation: https://youtu.be/JbWHa7KTeBg

Research Intern
May 2023 — Nov 2023

**Deezer Research**, Paris, France
*Self-led project on understanding diversity in music listening.*

Program Committee

**Chair in Music** at the upcoming ISMIR 2025

External Reviewer

Journals: *Topics in Cognitive Science*; *Scientific Reports*; *Transactions of the International Society of Music Information Retrieval (TISMIR)*; *Humanities and Social Sciences Communications*; *Evolutionary Human Sciences; Psychology of Music; Music & Science*

Co-founder

***aiar* collective**, bridging neuroscience with music performance — www.aiarcollective.com

## AWARDS & HONOURS

2025 **Best Student Paper Award**, Intl. Soc. of Music Information Retrieval (ISMIR)

2025 **Best Student Paper Award**, Intl. Conf. on Music Perception and Cognition (ICMPC)

2025 **Junior Research Fellowship**, King's College, University of Cambridge

2024 **Best Student Presentation Award**, Cultural Evolution Society Conf. (CES)

2023 **Best Paper Award**, Computational Humanities Research Conf. (CHR)

2023 **Nomination for Young Researcher Award,** Intl. Conf. on Music Perception and Cognition (ICMPC)

2023 **Travel Grant**, Intl. Conf. on Music Perception and Cognition (ICMPC) — $500

2022 **Scientific Research Fellow**, Diverse Intelligence Summer Institute (DISI) — $3,000

2022 **Art Grant** — Interactive Music Performance Using EEG, Leipzig City Council — €9,500

2021 **Nomination for Best Student Paper**, Intl. Soc. of Music Information Retrieval (ISMIR)

2021 **Nomination for Best Student Paper**, Intl. Conf. on Music Perception and Cognition (ICMPC)

2017 **Academic Excellence Scholarship**, Goldsmiths College — £3,000

2016 **Joe Brown Memorial Scholarship**, Music Department, Goldsmiths College — £500

## INVITED TALKS

2025 **Imagination Seminar Series, Sogang University**, S.Korea
*Computing cultural taste*

2025 **Neuroscience of Perception & Action Lab, Italian Institute of Technology**, Italy
*Computing cultural taste*

2025 **Whitehead Lecture Series, Goldsmiths, University of London**, UK
*Understanding music in the age of machine learning*

2025 **Faculty of Music, University of Cambridge**, UK
*Understanding music in the age of machine learning*

2024 **Music Cognition Lab, Princeton University**, US
*Global melody and rhythm reveal independent cultural clusters*

2024 **Music Department, Durham University**, UK
*Big data on music, implications and future outlooks*

2024 **AI Robotics Group, KIST Korea Institute of Science and Technology**, S. Korea
*Mechanisms of cultural spread and diversity through individual and population-level music patterns*

2024 **Cognitive Ethnomusicology Special Interest Group, Bates College**, US
*Global Pop: Studying the evolution of pop music across 51 countries since the 1950s*

2023 **Evolution & Social Cognition Group**, **Ecole Normale Supérieure**, France
*Understanding cultural diversity and spread through large-scale music listening behaviour*

2022 **CompMusic Lab, Keio University**, Japan
*Cross-cultural mood perception in pop songs and its alignment with algorithmic estimates*

2022 **Max Planck Institute for Evolutionary Anthropology**, Germany
*GlobalPOP: Studying the evolution of pop music across 51 countries since the 1950s*

2022 **Deezer Research**, France
*GlobalPOP: Investigating the historical trajectories of popular music around the world*

2022 **Brain Awareness Week Leipzig**, Germany
*Without a song or a dance what are we?: Origins of music and how our musical perception is shaped by culture*

## CONFERENCES

2025 **Oral** — Intl. Soc. for Music Information Retrieval Conf. (ISMIR), Dajeon, S.Korea
2025 **Oral** — Intl. Conf. on Music Perception and Cognition (ICMPC), Sao Paulo, Brazil
2025 **Poster** —Cognitive Science Society Annual Meeting (CogSci), San Francisco, US
2025 **Oral** — Explaining culture: an interdisciplinary approach, ENS, Paris, France
2024 **Oral** — Cultural Evolution Society (CES), Durham, UK
2024 **Oral** — Intl. Asso. of Empirical Aesthetics (IAEA), Mallorca, Spain
2023 **Oral** — Computational Humanities Research Conf. (CHR), Paris, France
2023 **Oral** — Intl. Conf. on Music Perception and Cognition (ICMPC), Tokyo, Japan
2022 **Oral** — Cultural Evolution Society (CES), Aarhus, Denmark
2022 **Oral** — Systematic Musicology (SysMus), Ghent, Belgium
2022 **Oral & Poster** — Cognitive Science Society Annual Meeting (CogSci), Toronto, Canada
2021 **Oral** — Intl. Soc. for Music Information Retrieval Conf. (ISMIR), Virtual
2021 **Oral** — Intl. Conf. on Music Perception and Cognition (ICMPC), Virtual
2020 **Poster** — Timbre 2020, Virtual
2019 **Oral** — Conf. on Interdisciplinary Musicology Embodiment Music, Graz, Austria
2019 **Poster** — Forum IRCAM Annual Workshop, Paris, France

## PRESS HIGHLIGHTS

2024 **New Scientist** — ‘Pythagoras was wrong about the maths behind pleasant music’.
https://www.newscientist.com/article/2419442-pythagoras-was-wrong-about-the-maths-behind-pleasant-music

2023 **BBC** — ‘Have your say: Does the weather influence the music you listen to?’.
https://www.bbc.co.uk/newsround/65457387

2023 **PhysOrg** — ‘New research shows how cultural transmission shapes the evolution of music’
https://phys.org/news/2023-03-cultural-transmission-evolution-music.html

## TEACHING

2026 *planned* **Faculty of Music, University of Cambridge**
Postgraduate seminar course on 'Global Musical Tastes in the Age of Big Data'

2025 **Tutorial, International Society for Music Information Retrieval Conference (ISMIR)**
3-hour tutorial on 'PsyNet: Collecting Large-Scale Behavioral Datasets for MIR Applications'

2024 **Department of Sociology, Seoul National University**
A 2-hour class on computational social science provided to Master's students

2021 **Department of Life Science, Leipzig University**
Seminar on deploying massive online experiments using PsyNet

## SKILLS

Language & Software R, Python, Javascript, HTML, SQL, Google BigQuery, AWS, HPC Computing, Gephi, Max MSP

Development Author of **tptR** R package (https://github.com/harin-git/tptR)
Psychoacoustic experiments run on R to determine interindividual differences in timbre perception

Contributed to **PsyNet** (www.psynet.dev)
A platform for running large-scale online behavioural experiments enabling complex paradigms

Open Science I publish all my working analysis code and data on OSF (https://osf.io/ecrhs) and GitHub (https://github.com/harin-git)

## REFERENCES

**Dr Nori Jacoby** (kj338@cornell.edu)
Assistant Professor, Department of Psychology, Cornell University

**Dr Marc Schönwiesner** (marcs@uni-leipzig.de)
Dean, Faculty of Life Sciences, Leipzig University

**Dr Minsu Park** (minsu.park@nyu.edu)
Assistant Professor, Social Research and Public Policy, NYU Abu Dhabi

## RANDOM FACTS

I've played on the stage of Berghain (Berlin), where I live monitored the brain activities of producer 'EgreEgore' and fed this back into the mix to manipulate the soundscapes and the underlying tempo of the beat. I also released two electronic pop music albums, which are available on major platforms ('Youth' by Harin and 'In Dreams' by Computer Hope).